Anushka Das

Kolkata, India — anushkadas05das@gmail.com — Portfolio — LinkedIn — GitHub

EXPERIENCE

Amazon ML Summer School (MLSS) Scholar — *Amazon* Jul 2026 – Present

- Selected as one of the scholars for Amazon ML Summer School 2026, a cohort-based program delivered directly by Amazon research scientists and applied ML researchers.
- Learning advanced machine learning topics including Supervised Learning, Deep Neural Networks, Dimensionality Reduction, Unsupervised Learning, Sequential Learning, Reinforcement Learning, Generative AI & LLMs, and Causal Inference through live sessions and hands-on modules.

Data Analytics Intern — *Oasis Infobyte (AICTE OIB-SIP) (Remote)* Jul 2026 – Aug 2026

- Architected Exploratory Data Analysis (EDA) workflows to extract actionable business insights, generating 7+ interactive visualizations utilizing Seaborn and Matplotlib to map customer demographics and seasonal sales trends.
- Implemented 3 end-to-end analytical solutions ranging from linear regression for price prediction to sentiment analysis—optimizing model cross-validation scores by 15 percent using Scikit-learn.

Blockchain Risk Developer — *Zetheta Algorithms Private Limited (Remote)* Jan 2026 – Feb 2026

- Engineered core decentralized infrastructure for the WorkBridge platform, integrating Web3 wallets to facilitate a secure, low-latency Crypto-Fiat Bridge.
- Developed an algorithmic market-making engine using the Avellaneda-Stoikov model to optimize bid-ask spreads and manage inventory risk in decentralized exchanges.
- Implemented an Avellaneda-Stoikov market-making engine to programmatically manage inventory risk, optimizing bid-ask spreads and enhancing high-frequency liquidity provision.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, HTML5, CSS3
- **AI & Machine Learning:** PyTorch, Scikit-learn, Pandas, NumPy, XGBoost, QLoRA, Deep Learning, RAG
- **Frameworks & Tools:** React, FastAPI, Flask, AWS, Docker, Git/GitHub, Jupyter, Streamlit, OpenCV

RESEARCH & PUBLICATIONS

Architectural Paradigms for Sovereign AI (Published on SSRN) — *Independent Research* Link

- Authored research mitigating API fragility by deploying Small Language Models (SLMs) via 4-bit NormalFloat (NF4) quantization, Native 1.58-bit ternary architectures, and Low-Rank Adaptation (LoRA).
- Demonstrated 93% inference OpEx reduction for 7B–14B parameter models with a TCO break-even at 450M tokens vs. proprietary cloud APIs.

PROJECTS

DeepfakeDetector: 2D FFT AI Inference Engine — *Python, PyTorch, React* GitHub Repo

- Architected a frequency-domain AI inference engine utilizing 2D FFT and ResNet-18, isolating synthetic media artifacts to achieve 99.12% detection accuracy.
- Engineered a distributed microservices stack using React and FastAPI across Vercel and Hugging Face, optimizing payload routing for $<$1.8s latency.
- Published research on a custom Gaussian Micro-blur preprocessing pipeline that neutralizes sensor noise, improving model robustness against diffusion generators.

Deweathering Engine — *React, FastAPI, OpenCV, RPCA* GitHub Repo

- Implemented a full-stack document restoration tool using Robust Principal Component Analysis to decompose degraded scans into low-rank text and sparse noise matrices.
- Designed a custom Adaptive Regularization heuristic achieving 95% automated parameter tuning, delivering end-to-end document restoration in $<$2.5 seconds.

Federated Fresh: Hybrid AI Orchestration Terminal — *FastAPI, Gemini, ChromaDB* GitHub Repo

- Designed a scalable AI orchestration terminal dynamically routing between local ChromaDB vector retrieval, live web sweeps, and LLM inference to maximize accuracy.
- Reduced local memory overhead by 75% by offloading high-latency text embeddings to the cloud, enabling stable deployment on constrained hardware.
- Engineered an asynchronous RAG pipeline leveraging FastAPI BackgroundTasks, achieving zero-lag UI performance during parallel chunking and vector injection.

Customer Churn Prediction Model — *Python, XGBoost, Scikit-learn, Streamlit* GitHub Repo

- Developed an end-to-end machine learning pipeline predicting customer churn, engineering custom behavioral features to achieve a 67% recall rate and 0.82 AUC.
- Deployed an interactive Streamlit web app, allowing stakeholders to perform real-time churn risk analysis via dynamic parameters.

EDUCATION

Techno Main Salt Lake — *B.Tech in Information Technology* 2023 – 2027

MEMBERSHIPS

- Member of **Google's Women Techmakers** & the **IEEE Robotics and Automation Technical Committee** on Machine Learning for Automation.

ACHIEVEMENTS & CERTIFICATIONS

- **Amazon MLSS 2026:** Selected as a Scholar for the Amazon ML Summer School 2026.
- **Hackathons:** SIH '25 Semi-Finalist (Top out of 250+ teams for project "Digi-Pramaan").
- **IndiaFirst Life (Bank of Baroda):** Unexpectedly shortlisted via Foundit and successfully cleared an interview for a Business Administration/Management role as a 6th-semester undergrad. Secured a 2-month training program with a stipend, leading to a full-time offer but respectfully declined it to focus on my 6th-semester studies.
- **Certifications:** Software Engineering (JPMorgan Chase), Data Analytics (Deloitte), Azure ML (Microsoft).